

ESP32 Smart Voltage Protector

FSMB Recruitment 2026 - Phase 1 | Embedded System Engineer

Project idea

I built a voltage and current protection demo using an ESP32. Two potentiometers act as simulated voltage and current sensors. When the reading goes above a set limit, the system shows a fault and turns off the green load indicator.

Hardware used

ESP32 (30-pin), two potentiometers, 16x2 I2C LCD (address 0x27), three pushbuttons, and two LEDs. The LCD shows the readings and current state. The green LED represents the load, while the red LED shows a fault.

How the system works

I used a Finite State Machine with four states: INIT, NORMAL, TRIPPED, and RESET_READY. In NORMAL, the green LED stays on. If voltage or current crosses its limit, the system enters TRIPPED and keeps the load indicator off. After the readings stay safe for one second, it enters RESET_READY. The user can then reset it using the MODE button or the web dashboard.

The default limits are 250 V and 8 A (simulated values). The buttons can change these limits locally. The ESP32 also creates a Wi-Fi network and hosts a simple dashboard to show live readings, update limits, and reset faults.

FSM diagram



Testing

I tested the potentiometers, LCD, LEDs, pushbuttons, and Wi-Fi dashboard on my ESP32. The displayed readings change when I turn the potentiometers. Overvoltage and overcurrent make the system trip. The fault remains until the readings are safe and I reset it. I can also change the limits from the buttons and the dashboard.

Limitations

This is a low-voltage demo, not a real mains protector. The potentiometers simulate electrical readings, and the LEDs only indicate the load state. No AC mains or real appliance is connected.